

6.6

Let X, Y be jointly continuous with the joint probability density function:

$$f(x, y) = xe^{-x(1+y)} \text{ for } x, y > 0$$

a

Find the marginal density function of X and Y

```
from sympy import Symbol, integrate
from scipy.integrate import dblquad
import math

x = Symbol("x")
y = Symbol("y")

print(integrate(x*math.e**(-x-x*y), y))
```

$$-1.0*2.71828182845905**(-x*y - x)$$

$$\begin{aligned} f_X(x) &= \int_0^\infty xe^{-x-xy} dy \\ &\rightarrow e^{-x} \int_0^\infty xe^{-xy} dy \\ &\rightarrow e^{-x} (-e^{-xy}|_0^\infty) \\ &\rightarrow e^{-x} (0 - (-e^{-x})) = e^{-2x} \end{aligned}$$

```
print(integrate(x*math.e**(-5*x), x))
```

$$1.0*(-0.2*x - 0.04)/2.71828182845905**(5*x)$$

The package didn't want to integrate with $y + 1$ so i swapped that for 5

$$\begin{aligned} f_Y(x) &= \int_0^\infty xe^{-x-xy} dx \\ f_Y(x) &= (-\frac{x}{(y+1)} - \frac{1}{(y+1)^2})e^{-x(y+1)}|_0^\infty \\ f_Y(x) &= (-\frac{x(y+1)+1}{(y+1)^2})e^{-x(y+1)}|_0^\infty \\ 0 - \frac{0(y+1)+1}{(y+1)^2} &= \frac{1}{(y+1)^2} \end{aligned}$$

b

Calculate $E(XY)$

$$E(XY) = \int_0^\infty \int_0^\infty x^2 y e^{-x(y+1)} dy dx$$

$$\rightarrow \int_0^\infty (x^2 e^{-x}) \frac{e^{-xy}}{x^2} \Big|_0^\infty dx$$

Note I removed the uv term as it goes to 0 with the bound integral

$$\rightarrow \int_0^\infty e^{-x-xy} \Big|_0^\infty dx$$

$$\rightarrow \int_0^\infty e^{-x} dx$$

$$\rightarrow -e^{-x} \Big|_0^\infty = 1$$

c

Calculate $E(\frac{X}{1+Y})$

$$E(\frac{X}{1+Y}) = \int_0^\infty \int_0^\infty \frac{x^2}{1+y} e^{-x(y+1)} dx dy$$

Integral calculator:

$$\int_0^\infty \frac{2}{(y+1)^4} dy$$

Integral calculator

$$\frac{-2}{3(y+1)^3} \Big|_0^\infty = \frac{2}{3}$$

6.7

Consider the triangle with vertices $(0,0), (1,0), (0,1)$. Suppose that (X,Y) is a uniformly chosen random point from this triangle

Since the area of the triangle is $\frac{1}{2}$, $f(x,y) = 2$ if $(x,y) \in$ the Triangle

a

Find the marginal density functions of X and Y

$$f_X(x) = \int_0^{1-x} 2 dx = 2(1-x)$$

$$f_Y(y) = \int_0^{1-y} 2 dy = 2(1-y)$$

b

Calculate $E(X)$ and $E(Y)$

$$E(X) = \int_0^1 \int_0^{1-x} 2x dy dx = \int_0^1 2x - x^2 dx = x^2 - \frac{1}{3}x^3 \Big|_0^1 = \frac{2}{3}$$

$$E(Y) = \int_0^1 \int_0^{1-y} 2y dx dy = \int_0^1 2y - y^2 dy = y^2 - \frac{1}{3}y^3 \Big|_0^1 = \frac{2}{3}$$

c

Calculate $E(XY)$

$$E(XY) = \int_0^1 \int_0^{1-x} 2xy dy dx = \int_0^1 x(1-x)^2 dx = \int_0^1 x - 2x^2 + x^3 \frac{1}{2}x^2 - \frac{2}{3}x^3 + \frac{1}{4}x^4 \Big|_0^1 = \frac{1}{12}$$

6.11

Let X have the density function $f_X(x) = 2x$ for $0 < x < 1$ and 0 else

Let Y be uniformly distributed on the interval $1, 2$

Assume X, Y are independent

a

Give the joint probability function of (X, Y)

$$f(x, y) = f_X(x)f_Y(y)$$

$$f_Y(y) = 1 \text{ for } y \in (1, 2)$$

$$f(x, y) = 2x \text{ for } x \in (0, 1) \wedge y \in (1, 2) \text{ and } 0 \text{ else}$$

b

Calculate the probability $P(Y - X \geq \frac{3}{2})$

$$2 - \frac{3}{2} = \frac{1}{2} \text{ and } 1 - \frac{3}{2} = -\frac{1}{2} \text{ so } P(Y - X \geq \frac{3}{2}) = \int_0^{\frac{1}{2}} \int_{x+\frac{3}{2}}^2 2xdydx$$

$$\rightarrow \int_0^{\frac{1}{2}} 4x - (2x^2 + 3x)dx$$

$$\rightarrow \int_0^{\frac{1}{2}} x - 2x^2 dx$$

$$\rightarrow \frac{1}{2}x^2 - \frac{2}{3}x^3 \Big|_0^{\frac{1}{2}}$$

$$\rightarrow \frac{1}{8} - \frac{1}{12} = \frac{1}{24}$$

6.18

Suppose that X and Y are integer valued random variables with joint probability mass function given by

$$p_{X,Y}(a,b) = \frac{1}{4a} \text{ for } 1 \leq b \leq a \leq 4$$

(a)

Show that this is indeed a joint probability mass function.

Essentially, show $\sum \sum p(a,b) da db = 1$

$$\sum_{a=1}^4 \sum_{b=1}^a \frac{1}{4a} = \sum_{a=1}^4 \frac{1}{4a} a = \sum_{a=1}^4 \frac{1}{4} = 1$$

(b)

Find the marginal probability mass functions of X and Y.

for $a, b \in 1, 2, 3, 4$

$$p_X(a) = \sum_{b=1}^a \frac{1}{4a} = \frac{1}{4}, \forall a$$

$$p_Y(b) = \sum_{a=b}^4 \frac{1}{4a} = \frac{1}{4} \sum_{a=b}^4 \frac{1}{a}$$

$$p_Y(1) = \frac{1}{4} \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) = \frac{1}{4} \frac{25}{12} = \frac{25}{48}$$

$$p_Y(2) = \frac{1}{4} \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) = \frac{1}{4} \frac{13}{12} = \frac{13}{48}$$

$$p_Y(3) = \frac{1}{4} \left(\frac{1}{3} + \frac{1}{4} \right) = \frac{1}{4} \frac{7}{12} = \frac{7}{48}$$

$$p_Y(4) = \frac{1}{4}(\frac{1}{4}) = \frac{1}{16}$$

(c)

Find $P(X = Y + 1)$

$$\sum_{a=2}^4 p_{X,Y}(a, a-1) = \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \frac{13}{48}$$